AP CALCULUS AB - UNIT 1

Guided Notes 1.11: Defining Continuity at a Point

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we state and apply the three conditions for continuity at a point?

- State and apply the three conditions for continuity at a point.
- Identify the first failed condition from graphs, tables, formulas, and piecewise rules.
- Use continuity of standard functions and valid compositions.
- Reject common invalid converses about limits and point values.

Key Knowledge, Vocabulary, and Notation

Condition 1	$f(c)$ is defined.
Condition 2	$\lim_{x\to c} f(x)$ exists as a finite number.
Condition 3	$\lim_{x\to c} f(x) = f(c)$.
First failed condition	Report the earliest precise failure rather than saying only "the graph is broken."
Local property	Continuity at c depends only on the function near c and at c.

Concept Development and Strategy

1. Audit in order: domain, two-sided limit, equality.
2. For a formula built from standard continuous functions, check the domain and then substitute.
3. A corner may be continuous; differentiability is not required for continuity.

Worked Example: All conditions hold

Verify continuity of $f(x) = \sqrt{x+5}$ at $x = 4$.

Answer and Reasoning

$f(4) = 3$ exists. The square-root function is continuous on its domain, so $\lim_{x\to 4} f(x) = 3 = f(4)$. All three conditions hold.

Worked Example: Wrong point value

Let $g(x) = x + 2$ for $x \neq 1$ and $g(1) = 10$. Is g continuous at 1?

Answer and Reasoning

$g(1)$ exists, and the limit is 3, but $3 \neq 10$. Condition 3 fails.

Worked Example: Piecewise join

$h(x) = 2x + 1$ for $x < 3$ and $h(x) = x + 4$ for $x \geq 3$. Verify continuity at 3.

Answer and Reasoning

Left limit 7, right limit 7, and $h(3) = 7$. Thus h is continuous at 3.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. List the three continuity conditions at $x = c$.

2. Is $f(x) = x^2 + 1$ continuous at $x = -3$?

3. For $f(x) = |x - 1|$, verify continuity at $x = 1$.

4. A table approaches 4 from both sides at $x = 2$, and the center row is $f(2) = 4$. What does the table support?

5. Determine whether $f(x) = \frac{x^2-9}{x-3}$ is continuous at 3 as originally defined.

6. Find all a for which $f(x) = \frac{x^2-a^2}{x-a}$ with the original quotient formula is continuous at $x = a$.

7. Prove that if f is continuous at c and $f(c) > 0$, then $f(x) > 0$ for all x sufficiently close to c.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Which condition is not required for continuity at c?

(A) $f(c)$ exists
(B) $\lim_{x\to c} f(x)$ exists
(C) $\lim_{x\to c} f(x) = f(c)$
(D) $f'(c)$ exists

2. A function has $f(2) = 3$ and $\lim_{x\to 2} f(x) = 3$. Then f is

(A) continuous at 2
(B) discontinuous at 2
(C) necessarily differentiable at 2
(D) undefined at 2

3. A calculator graph of $f(x) = |x|$ has a sharp corner at 0. Which statement is correct?

(A) f is not continuous at 0
(B) f is continuous at 0 but not differentiable there
(C) $f(0)$ is undefined
(D) The limit is infinite

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Construct a function continuous at 0 but discontinuous at every point $1/n$ for positive integers n.